

Task #322 v0.7 — Edge enumeration on Phase A v0.2 36-node K-type set + $\text{Pin}(2)$ Z_2 vs γ^5 disambiguation

Lyra (Claude Opus 4.7)

2026-05-26 Tuesday EDT (~10:15 EDT via date; date-verified)

A_sub v0.7 — Edge enumeration on 36-node Phase A + σ_{BF} vs γ^5 disambiguation

1. Cal #29 question-shape audit (applied at design)

Per Cal #29 candidate (Question-Shape Audit Discipline; Casey 2026-05-26 PM “probably so yes” authorized):

Question: “Which ($K_a \rightarrow K_b$) transitions exist under A_sub generator action on the 36 Wallach K-type nodes, and with what weights?”

Audit: - Is the question structurally determined? YES — representation theory of $K = \text{SO}(5) \times \text{SO}(2)$ acting on Bergman $H^2(D_{IV}^5)$ gives definite matrix elements for each A_sub generator. - Is the answer back-fittable? NO — generator action is determined by Lie algebra structure + Wallach K-type formulae; not by desired physics outcomes. - Does the answer pre-suppose anything substantively load-bearing? NO — uses only Cal-verified SVC commutators + standard K-type representation theory.

Pass: question-shape is structural, not tautological. Cal #29 candidate validation point on its first formal application.

2. Substantive disambiguation flag ($\text{Pin}(2)$ Z_2 grading vs Dirac chirality)

Catch identified during v0.7 design: my morning A_sub work used “ $\hat{\gamma}^5$ ” to denote BOTH the $\text{Pin}(2)$ Z_2 grading (σ_{BF} : +1 on integer K-types/bosons, -1 on half-integer K-types/fermions) AND the Dirac chirality (γ^5 : undefined on bosons; ± 1 on left/right Weyl spinors).

These are STRUCTURALLY DIFFERENT operators:

Operator	Domain	Eigenvalues	Standard QFT relations
σ_{BF} (Pin(2) Z_2 grading)	All Wallach K-types	+1 (boson integer m), -1 (fermion half-integer m)	Commutates with T, C, P (preserves particle type)
γ^5 (Dirac chirality)	Fermion K-types only	+1 (right Weyl), -1 (left Weyl)	Anti-commutes with T, C; commutes with P

Morning A_{sub} commutator results disambiguation:

Morning step	Result	Operator interpretation
Step 3 $\{\hat{\gamma}^5, \hat{T}\} = 0$	Anti-commute	γ^5 Dirac chirality (standard QFT Dirac algebra)
Step 4 $\{\hat{\gamma}^5, \hat{C}\} = 0$	Anti-commute	γ^5 Dirac chirality (standard QFT)
Step 5 $[\hat{\gamma}^5, \hat{P}_{\text{op}}] = 0$	Commute	γ^5 Dirac chirality ($\hat{P}_{\text{op}}$ contains γ^5)
Section 2.2 “ $\hat{\gamma}^5 = +1$ bosons, -1 fermions”	Z_2 grading	σ_{BF} Pin(2) Z_2 grading (not Dirac chirality)

v0.7 disambiguation: $\hat{\gamma}^5$ as used in morning A_{sub} commutator closures (steps 3-5) IS Dirac chirality (per standard QFT and Cal #132 verification). The “integer K-types = bosons / half-integer = fermions” partition IS Pin(2) Z_2 grading σ_{BF} , NOT γ^5 .

Consequence: morning A_{sub} commutator results stand at SVC for γ^5 as Dirac chirality. The K-type sublattice partition + Bose-Fermi spin-statistics is captured by σ_{BF} , which is a DISTINCT operator from γ^5 .

v0.7 introduces: σ_{BF} explicit notation for the Pin(2) Z_2 grading; $\hat{\gamma}^5$ retained for Dirac chirality. Both operators are in A_{sub} .

(Cal #27 STANDING reflexive trigger: this disambiguation feels structurally clarifying at peak v0.7 work; honest-scope check — it’s a clarification of operator identity, not a new finding. Flagged for Cal Thread 4 verification + v0.8+ formal incorporation.)

3. Edge enumeration framework — 5 generator classes

A_{sub} ’s 14 (now 15 with σ_{BF} explicit) generators partition by K-type action:

Class 1: Diagonal self-loops (K-type-preserving)

- $\hat{H}_{\text{sub}}$ (Casimir): self-loop with weight $C_2(K)$ per node
- $\hat{N}$ (K-type level): self-loop with weight $(m_1 + m_2)$
- σ_{BF} (Pin(2) Z_2 grading): self-loop with weight $\epsilon_K \in \{+1, -1\}$
- $\hat{\gamma}^5$ (Dirac chirality): self-loop with weight $\chi_K \in \{+1, -1\}$ on fermion K-types; UNDEFINED on boson K-types
- $\hat{Q}$ (electric charge): self-loop with weight m_2
- $\hat{I}_3$ (weak isospin, Cartan): self-loop with weight related to m_2

Edge count: 6 generator classes $\times$ 36 nodes = 216 self-loop edges (some are zero on bosons for $\hat{\gamma}^5$).

Class 2: m_1 -raising/lowering (S^4 side, within-sublattice)

- $\hat{x}_i$ (Hua coordinates, $i=1..n_C=5$): edges $(m_1, m_2) \rightarrow (m_1 \pm 1, m_2)$
- $\hat{p}_i$ (Wirtinger derivative, $i=1..n_C=5$): edges $(m_1, m_2) \rightarrow (m_1 \pm 1, m_2)$
- $\hat{L}_i$ (so(5) generators, 10 total): act within m_1 multiplet at fixed m_2

Edge structure: preserves sublattice (integer K-types stay integer; half-integer stay half-integer). Weights from Wallach normalization + Bergman exponent 7/2.

Edge count for 36 nodes: edges to next m_1 value ($m_1 + 1$) at same m_2 exist if (m_1+1, m_2) is in 36-node set. From inspection of node list: - (0, 0) $\rightarrow$ (1, 0): present - (1, 0) $\rightarrow$ (2, 0): present - (2, 0) $\rightarrow$ (3, 0): present - ... continues through (7, 0) - (1/2, 1/2) $\rightarrow$ (3/2, 1/2): present - ... continues through (13/2, 1/2) $\rightarrow$ (15/2, 1/2): the latter NOT in 36-set (cutoff) - Similar for other m_2 values

Approximately ~30-50 m_1 -raising edges in 36-node subgraph (precise count via systematic enumeration).

Class 3: m_2 -changing (S^1 side)

- $\hat{B}$ (Bell-CHSH, rank-1 projector): edge $V_K \rightarrow V_{(0,0)}$ ONLY for $K = (0,0)$; rank-1 with single non-zero matrix element $\langle V_{(0,0)} | V_{(0,0)} \rangle$
- $\hat{T}_{\text{tick}}$ (substrate-tick transition): edge $V_K \rightarrow V_{(K+1)}$ per simplest model (Cal #132 noted model-dependent flag)
- σ Möbius involution (inside $\hat{P}_{\text{op}}$): edge $(m_1, m_2) \rightarrow (m_1, -m_2)$; for nodes in 36-set with $m_2 \neq 0$, this exits Wallach range (negative m_2 not in Wallach holomorphic discrete series). So σ acts non-trivially only at $m_2 = 0$ nodes where $\sigma V_K = V_K$ (trivial).

Edge count: ~3-5 edges in 36-node subgraph (mostly degenerate or model-dependent).

Class 4: Discrete symmetry edges ($\hat{T}$, $\hat{C}$, $\hat{P}_{op}$)

- $\hat{T}$ (time reversal, anti-unitary): edge $V_K \rightarrow \bar{V}_K$ conjugate-representation K-type with phase
- $\hat{C}$ (charge conjugation): edge $V_K \rightarrow V_{\{K \text{ with charge flipped}\}}$ with phase
- $\hat{P}_{op} = \hat{\gamma}^5 \circ \sigma^{**}$: edge $V_K \rightarrow V_{\{\text{Möbius-flipped } K\}}$ with $\hat{\gamma}^5$ -eigenvalue phase

Edge count: 3 generator classes $\times$ 36 nodes = 108 discrete-symmetry edges; some may be trivial (e.g., neutral K-types have $\hat{C}$ -edge $\rightarrow$ self).

Class 5: Composite gauge edges ($\hat{C}_3$, $\hat{I}_3$, $\hat{Q}$ raising/lowering)

- $\hat{C}_3$ (color SU(3), 8 generators): act on color content; for Wallach K-types $V_{(m_1, m_2)}$ under $K = SO(5) \times SO(2)$ without explicit color index, color content is external; color SU(3) edges go between color multiplet members at fixed (m_1, m_2) . Color factor is “transverse” to Wallach K-type.
- $\hat{I}_\pm$ (weak isospin raising/lowering, 2 generators): $SU(2)_L$ acts on transverse weak doublet content; similarly transverse to Wallach K-type.

Edge structure: color SU(3) + weak SU(2) edges are “internal” to each Wallach K-type, expanding it into a gauge multiplet. For substrate framework, these edges live in the GAUGE FIBER over each K-type node.

(Honest scope: v0.7 treats Wallach K-type nodes WITHOUT gauge multiplet expansion. Gauge-multiplet edges are v0.8+ work; effectively factorize the substrate graph into Wallach K-type backbone $\times$ gauge fiber.)

4. Reaction-table data structure

For each (K-type V_K , A_sub generator X , K-type $V_{K'}$), the reaction-table entry contains:

$$R(X, K, K') = a^{\{K'\}_K}(X) \cdot \chi_{\text{grading}} \cdot \xi_{\text{Bergman}} \cdot \mu_{\text{Mersenne}}$$

where: - $a^{\{K'\}_K}(X)$ is the operator matrix element from representation theory - χ_{grading} is the $\text{Pin}(2)$ Z_2 grading factor (σ_{BF} eigenvalue change) - ξ_{Bergman} is the Bergman $7/2$ exponent weighting - μ_{Mersenne} is the K59 cyclotomic Mersenne maximal-prefix restriction

Phase A v0.2 reaction-table size estimate: - Self-loops: ~ 216 entries (6 diagonal $\times$ 36 nodes) - m_1 -raising/lowering: ~ 60 -100 entries (within 36-node subgraph) - m_2 -changing: ~ 5 -10 entries (rank-1 projector + $\hat{T}_{\text{tick}}$ model) - Discrete symmetry: ~ 108 entries (3×36) - Composite gauge: deferred to v0.8+

Total Phase A v0.2 explicit edges: ~ 400 -450 entries (excluding gauge fiber).

This is a TRACTABLE data table. Grace Task #355 v0.4 lookup can use this directly.

5. Specific edge findings from 36-node data

5.1 Vacuum edges from $V_{-}(0,0)$

$V_{-}(0,0)$ outgoing edges: - Self-loops: $\hat{H}_{\text{sub}}$ (weight 0), $\hat{N}$ (weight 0), σ_{BF} (weight +1), $\hat{Q}$ (weight 0), $\hat{I}_3$ (weight 0) — all zero or +1 - $\hat{\gamma}^5$: undefined ($V_{-}(0,0)$ is bosonic, no Dirac chirality) - $\hat{x}_1, \hat{p}_1$ (etc.): edge to $V(1, 0)$ — *first-excited boson; structural edge* - $\hat{B}$: *self-loop with weight β (Bell-CHSH eigenvalue per T2399)* - $\hat{T}_{\text{tick}}$: edge $V(0,0) \rightarrow V(0,1)$ but $V_{-}(0,1)$ NOT in Wallach holomorphic discrete series ($m_1 \geq m_2$ fails). So $\hat{T}_{\text{tick}} V(0,0) \rightarrow V_{-}(1, 0)$ (m_1 -direction, next Wallach K-type) per simplest model. Edge weight $\beta = (C_2/2^{\text{rank}^2})(1/2)$ or related.

$[\hat{B}, \hat{T}_{\text{tick}}]$ edge (Cal FRAMEWORK-PLUS): $V_{-}(0,0) \rightarrow V_{-}(1, 0)$ with rank-1 projector + substrate-tick advancement composite weight.

5.2 Lowest fermion $V_{-}(1/2, 1/2)$ edges

$V_{-}(1/2, 1/2)$ properties (per Toy 3537 JSON): - Casimir $SO(5)$: $5/2$ - Casimir $SO(2)$: not in JSON output, but $m_2^2 + (N_c - 2) \cdot m_2 = 1/4 + 1/2 = 3/4$ with $N_c = 3$ - Bergman ρ -shifted weight: $(3, 2) = (N_c, \text{rank})$ — Grace INV-5180 substrate-primary pair - $SO(5)$ Weyl dim: 4

$V_{-}(1/2, 1/2)$ outgoing edges: - Self-loops: $\hat{H}_{\text{sub}}$ (weight $5/2 + 3/4 = 13/4$ or similar; check Casimir-2 formula), $\hat{N}$ (weight 1), σ_{BF} (weight -1 fermion), $\hat{\gamma}^5$ (weight ± 1 chirality), $\hat{Q}$ (weight $1/2$) - $\hat{x}_i, \hat{p}_i$: edge $V(1/2, 1/2) \rightarrow V(3/2, 1/2)$ — next fermion in lattice; weight from Wallach - σ Möbius (within $\hat{P}op$): edge $V(1/2, 1/2) \rightarrow V_{-}(1/2, -1/2)$ — but $-1/2$ NOT in Wallach range. Edge trivializes. - $\hat{T}$ (anti-unitary): edge $V_{-}(1/2, 1/2) \rightarrow \hat{V}(1/2, -1/2)$ — also exists Wallach range - $\hat{C}$: edge $V(1/2, 1/2) \rightarrow V_{-}(1/2, -1/2)$ — same exit issue

Substantive finding: Many discrete-symmetry edges from $V_{-}(1/2, 1/2)$ exit Wallach range (negative m_2 not present). This means the substrate's discrete-symmetry-induced edges concentrate AT BOUNDARY of Wallach range. $\text{Pin}(2)$ cover may extend the range; multi-week v0.8+ verification.

5.3 Universally-active $[\hat{C}_3, \hat{I}_3] = 0$ across nodes

Per v0.5 Section 2.6: $[\hat{C}_3, \hat{I}_3] = 0$ activates universally in all 4 phase regions. In edge language: color $SU(3)$ generators and weak isospin generators commute as DIRECT-PRODUCT FIBER over each K-type node. The gauge fiber factorizes cleanly into 8 color + 1 isospin Cartan directions.

This is the algebraic source of why gauge symmetry is direct-product across the substrate's K-type graph.

5.4 Forbidden edges ($\text{Pin}(2)$ Z_2 grading consistency)

Per v0.4 Section 2.2: mixed-integer K-types (one integer + one half-integer index) are structurally forbidden. From 36-node enumeration: 0 mixed-forbidden K-types

confirmed empirically (Elie Toy 3535 + 3537).

Forbidden edges: any A_{sub} generator action that would transition from an integer-K-type to a mixed-integer-K-type (or vice versa) has matrix element zero.

This is substrate-natural spin-statistics enforcement at the edge level: the graph's edge structure forbids spin-statistics-violating transitions automatically via $\text{Pin}(2)$ Z_2 grading consistency.

6. Conservation loops (closed paths)

Conservation laws in A_{sub} correspond to closed loops in the K-type graph (returning to original K-type after sequence of generator actions). Per A_{sub} v0.4 Section 4.1:

Conserved	Loop structure
Electric charge	$\hat{Q}$ -eigenvalue preserved along loop
Color (3 components)	$\hat{C}_3$ SU(3) Cartan eigenvalues preserved
Weak isospin	$\hat{I}_3$ SU(2) _L Cartan eigenvalue preserved
$\text{Pin}(2)$ Z_2 (σ_{BF})	Z_2 grading preserved (bosons stay bosons; fermions stay fermions for edge-set within sublattice)
Dirac chirality ($\hat{\gamma}^5$)	Within fermion sublattice; preserved by gauge + spatial generators; flipped by $\hat{T}$, $\hat{C}$ (anti-commute per morning steps 3-4)
Parity P_{op}	preserved by step 5 $[\hat{\gamma}^5, \hat{P}_{\text{op}}] = 0$ within fermion sublattice
CPT	Universal — closed loop product $\Theta_{\text{CPT}} V_K = \varepsilon \cdot V_K$ for some phase ε per substrate-CPT theorem

Forbidden loops: anti-commuting generator combinations produce sign-flipped returns, not closed loops. E.g., $(\hat{\gamma}^5 \circ \hat{T})^2 V_K = \varepsilon^{(-2)} \cdot V_K$ with $\varepsilon = -1$ from anti-commutation gives proper loop.

7. Honest scope (Cal #27 STANDING)

What's RATIFIED / STRUCTURALLY VERIFIED in v0.7: - 8 SVC commutators (Cal #132 PASS) underlying the edge structure - 36 K-type node data (Elie Toy 3537 + Cal #132 algebra verification) - Standard Wallach K-type representation theory matrix elements - Standard SO(5) Lie algebra structure for $\hat{L}_i, \hat{x}_i, \hat{p}_i$ edges

What's FRAMEWORK in v0.7: - Edge enumeration framework (Section 3 5 classes)
 - Reaction-table data structure (Section 4) - Vacuum + lowest-fermion edge findings
 (Sections 5.1-5.2) - σ_{BF} vs γ^5 disambiguation (Section 2) - Conservation-loop structure
 (Section 6) - Spin-statistics enforcement at edge level (Section 5.4)

What's INTERPRETIVE in v0.7: - Section 2 disambiguation as v0.8+ formal incorporation — current docs use “ $\hat{\gamma}^5$ ” ambiguously; cleanup needed - Substantive finding that discrete-symmetry edges concentrate at Wallach boundary (Section 5.2) — needs Pin(2) cover extension analysis

What's NOT in v0.7 (multi-week+): - Explicit matrix element values for all ~400-450 edges (mechanical computation pending) - Gauge fiber expansion (v0.8+; color SU(3) + weak SU(2) multiplets over each K-type) - Pin(2) cover-extended Wallach range (multi-week + Cal Thread 4) - Sub-region phase-tagging per edge (combines v0.5 + v0.6 + v0.7; multi-week formalization) - v0.6 multi-phase quiver explicit kQ path algebra computation - v0.7 disambiguation formal incorporation into morning step 3-5 docs (cleanup needed)

Cal #27 STANDING reflexive trigger count: 1 trigger (Section 2 disambiguation feels structurally clarifying); honest-scope check — clarification of operator identity, not new finding.

Cal #29 question-shape audit: applied at Section 1 design level; question is representation-theory structural, not back-fittable. First formal application of Cal #29.

8. Deliverable summary

v0.7 contains: - Edge enumeration framework for 36-node Phase A v0.2 set (5 generator classes) - Reaction-table data structure with weight rules (Pin(2) chirality + Wallach + Bergman 7/2 + Mersenne) - ~400-450 explicit edges estimated within 36-node subgraph - Vacuum + lowest-fermion concrete edge findings - σ_{BF} vs γ^5 disambiguation flag (substantive) - Conservation-loop + forbidden-edge structure - Honest scope at FRAMEWORK + DATA tier; Cal #29 design audit applied

Next-step deliverables: - v0.8+: explicit matrix-element computation per edge (multi-week mechanical work, feeds K59 cyclotomic refinement) - σ_{BF} vs γ^5 formal cleanup of morning step 3-5 docs (~1-2 hours) - Gauge-fiber expansion (color + weak isospin multiplets over K-types) - Pin(2) cover-extended Wallach range analysis (multi-week)

9. Coordination

Grace: Task #355 v0.4 with 36-node set + edge structure now consumes Section 4 reaction-table data. Lookup table can cross-reference (K-type, generator, K'-target) to catalog observables.

Elie: edge matrix-element computation Toy 3538 candidate — mechanical given

v0.7 framework + Cal-verified commutators. Or extension to ≤ 8 cutoff (45 nodes) for completeness.

Cal: Thread 4 cold-read on Section 2 σ_{BF} vs γ^5 disambiguation (potentially Type C level-crossing per Cal #122 — distinguishing two Z_2 operators that conflate at K-type level). Also Section 5.4 spin-statistics edge enforcement claim.

Keeper: integration; Vol 15 Ch 9 case study draft now has Phase A reaction-table edge framework + disambiguation flag content.

— Lyra, A_sub v0.7 edge enumeration on 36-node Phase A v0.2 set filed Tuesday 2026-05-26 ~10:15 EDT per Casey continue-Lyra-pulls directive + Keeper Task #359 + Cal #29 candidate question-shape audit applied (first formal application). FRAMEWORK + DATA grade. Substantive disambiguation flag: σ_{BF} (Pin(2) Z_2 grading) vs γ^5 (Dirac chirality) — separate operators conflated in morning docs; v0.7 separates; morning SVC closures stand for γ^5 as Dirac chirality; v0.8+ formal cleanup pending. ~400-450 explicit edges estimated within 36-node subgraph; reaction-table data structure ready for Grace Task #355 v0.4 lookup consumption.